

SHRIDHAR KR. MATHUR

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Professional Summary

Final-year B.Tech. Computer Science (AI & ML) student at Abdul Kalam Technical University with hands-on experience building intelligent systems using Python, NLP, Computer Vision, and Speech Recognition. Designed and deployed end-to-end ML-powered desktop applications during an internship at NIELIT and through independent projects. Demonstrated leadership as Editorial Board Coordinator and Head of Sports at large-scale campus events. Seeking an AI/ML engineering or software development role to solve real-world problems at scale.

Professional Experience

National Institute of Electronics & Information Technology (NIELIT) 🌐 Lucknow, India
Machine Learning Intern — (Artificial Intelligence & Machine Learning Internship) Aug. 2025 – Sep. 2025

- Designed and implemented supervised & unsupervised ML pipelines in **Python**, applying algorithms including Decision Trees, Random Forest, K-Means, and SVM on datasets of 1,000+ records to solve real-world classification and clustering problems.
- Built a real-time Object Detection System UI with a **YOLO-based backend model**, enabling live-video and image inference across 20+ object classes with bounding-box overlays and confidence-score display.
- Evaluated model performance using accuracy, precision, recall, and F1-score; compiled structured evaluation reports and presented findings to senior mentors.
- Gained end-to-end exposure to the complete ML development lifecycle — from data preprocessing and feature engineering through model training, evaluation, and deployment-ready packaging.

Projects

Object Detection System Interface | *Python, OpenCV, YOLO, Tkinter/PyQt, NumPy* Aug. 2025 – Sep. 2025

- Architected a full-stack desktop interface for a real-time **Computer Vision** application, enabling users to process static images and live video streams through a **backend YOLO-based object detection model**.
- Implemented real-time bounding-box rendering, object-label overlays, and confidence-score display across 20+ object classes, enabling instant visual feedback from multiple input sources at 20+ FPS on standard hardware.
- Designed a modular, plug-and-play architecture allowing seamless backend model swapping (YOLOv5 → YOLOv8) with zero front-end code changes, significantly improving project extensibility.

EDITH — Personal AI Assistant (Ongoing) | *NLP, SpeechRecognition, OS Automation* Sep. 2025 – May. 2026

- Architected a fully voice-controlled AI assistant using **Python**, leveraging **NLP intent classification** to interpret natural language commands and automate system-level tasks — app launching, file management, and web navigation.
- Integrated Speech Recognition and text-to-speech synthesis (pyttsx3), achieving sub-2-second command response latency under standard operating conditions.
- Implemented an OS-level auto-launch mechanism via startup scripts, enabling zero-touch assistant activation on every system boot with no user intervention.
- Mapped voice inputs to 15+ system-automation routines using NLP intent-classification, with a modular registration system for iterative vocabulary expansion.
- In progress: Integrating an LLM-based conversational layer (OpenAI API / local LLM) to enable multi-turn dialogue and context-aware command interpretation.*

Technical Skills

Programming Languages: Python, C++, Java, HTML, CSS, JavaScript (Fundamentals)

AI / ML & Data Science: Machine Learning, Deep Learning (Fundamentals), Natural Language Processing (NLP), Computer Vision, Speech Recognition, NumPy, Pandas, OpenCV, TensorFlow (Fundamentals), PyTorch (Fundamentals)

Frameworks & Libraries: React.js, Angular, Node.js, SpeechRecognition, pyttsx3, PyAutoGUI

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Soft Skills: Leadership, Cross-functional Collaboration, Technical Communication, Critical Thinking, Creativity, Team Management

Education

Abdul Kalam Technical University (AKTU) Lucknow, Uttar Pradesh
B.Tech. — Computer Science (Artificial Intelligence & Machine Learning) Aug. 2022 – Present

- CGPA: [6.1 / 10.0] | Expected Graduation: **Oct 2026**

Certifications & Professional Development

- **Internship Certificate — AI & Machine Learning using Python**  *NIELIT, 2025*
- **Data Structures using Java — 12-Week Programme**  *Logicpro Infosystem & SMS Lucknow*
- **Front-End Development Bootcamp — 2-Day Intensive**  *Coding Blocks*
- **Fundamentals of C Programming**  *IIT Kanpur (Organised Certification)*

Achievements & Extracurricular Activities

- **Head of Sports — SPECTRUM 2024:** Directed all sports operations and volunteer coordination for a 3-day inter-college festival (Apr. 25–27, 2024), managing logistics for 100+ participants across multiple concurrent sporting events.
- **IBM SkillsBuild Workshop** (Nov. 20–25, 2024, 6 days): Completed intensive hands-on sessions on AI fundamentals, cloud computing, and enterprise software development practices delivered by IBM-certified instructors.
- **Live MERN Stack Bootcamp** (2024, 3-Day Intensive): Built a functional full-stack web application using MongoDB, Express.js, React.js, and Node.js; gained practical exposure to RESTful API design and component-based UI development.